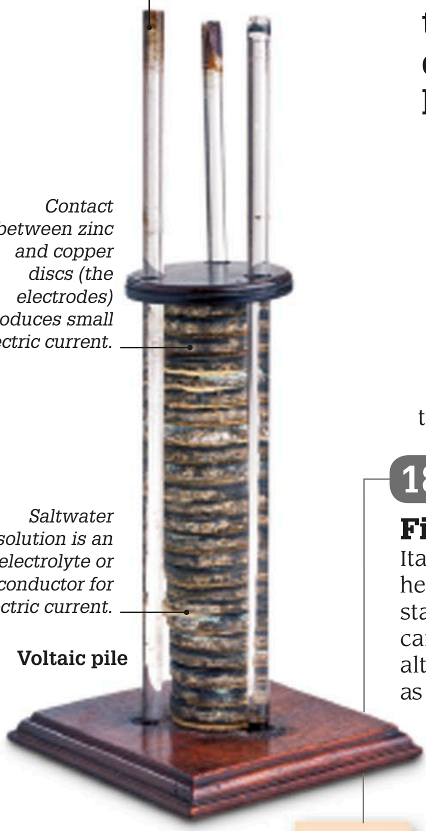


Voltaic pile

In 1801, Jean-Baptiste Lamarck came up with the term "invertebrates" to describe animals without backbones.

1804
First railroad trip
Richard Trevithick, a Cornish engineer, built a high-pressure steam engine and mounted it on wheels. His locomotive pulled five wagons carrying 70 passengers and 11 tons (10 metric tons) of coal a distance of 9 miles (14 km). It traveled at a speed of 5 mph (8 km/h).

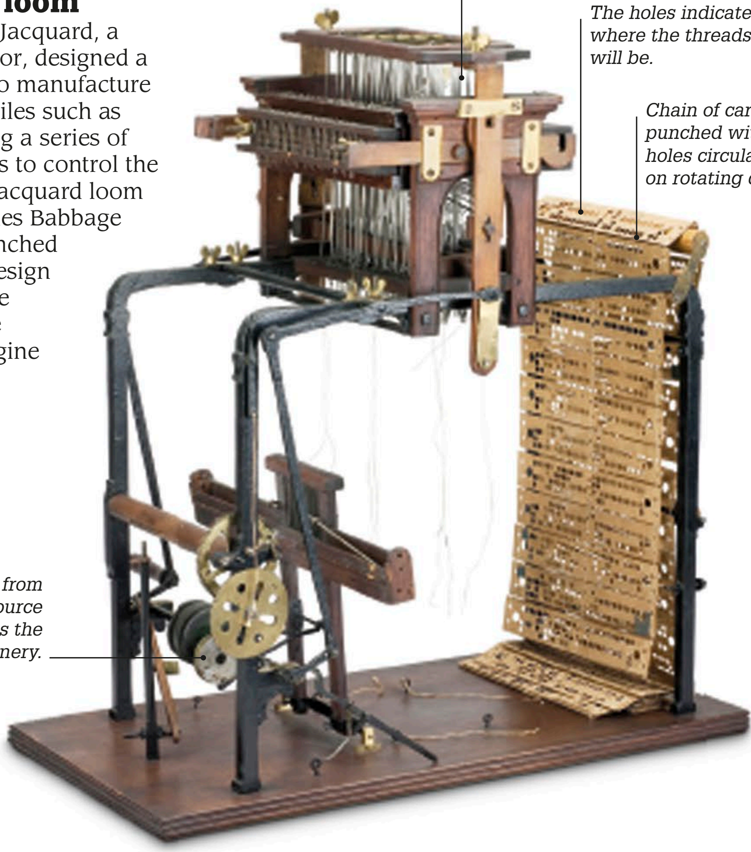
1800
First electric battery
Italian inventor Alessandro Volta found he could create an electric current by stacking discs of zinc, copper, and cardboard soaked in saltwater in alternating layers. His device, known as a "voltaic pile," was the first "wet cell" electric battery. Adding more discs increased the amount of electricity generated.


Gay-Lussac and Biot make their ascent

1804
Up in the air
French scientists Joseph Louis Gay-Lussac and Jean-Baptiste Biot ascended to a record height of 23,108 ft (7,016 m) in a hot-air balloon to study the composition of Earth's atmosphere at altitude.

1800 — 1805

1801
Jacquard loom
Joseph-Marie Jacquard, a French inventor, designed a power loom to manufacture elaborate textiles such as brocades using a series of punched cards to control the pattern. The Jacquard loom inspired Charles Babbage to employ punched cards in his design for a prototype computer, the Analytical Engine (see p.124).



1803
Atomic theory
In a lecture to an audience in Manchester, English chemist John Dalton suggested that all matter is composed of atoms, and that atoms of the same element are identical. He compiled a table of elements based on their atomic weights.

Dalton's table of elements